

//Q3

//<https://github.com/AkashChand6n/Devops-retest.git>

//Bash script that collects and displays the following system information: current date and time, CPU usage, memory usage, disk space usage, and network statistics. The script should also send an email alert if CPU or memory usage exceeds 80%.

```
#!/bin/bash
```

```
# Email settings
```

```
EMAIL="akashchand6n@gmail.com"
```

```
SUBJECT="System Resource Alert on $(hostname)"
```

```
# Collect system info
```

```
echo "==== System Information ====="
```

```
echo "Date and Time: $(date)"
```

```
echo
```

```
echo "==== CPU Usage ====="
```

```
CPU_IDLE=$(top -bn1 | grep "Cpu(s)" | awk '{print $8}')
```

```
CPU_USAGE=$(echo "100 - $CPU_IDLE" | bc)
```

```
echo "CPU Usage: $CPU_USAGE%"
```

```
echo
```

```
echo "==== Memory Usage ====="
```

```
MEM_TOTAL=$(free | awk '/Mem:/ {print $2}')
```

```
MEM_USED=$(free | awk '/Mem:/ {print $3}')
```

```
MEM_USAGE=$(echo "$MEM_USED * 100 / $MEM_TOTAL" | bc)
```

```
echo "Memory Usage: $MEM_USAGE%"
```

```
echo
```

```
echo "==== Disk Space Usage ====="
```

```
df -h
```

```
echo
```

```

echo "==== Network Statistics ====="

if command -v ifstat &> /dev/null; then
    ifstat 1 1
else
    echo "ifstat not found. Showing fallback from /proc/net/dev:"
    cat /proc/net/dev | awk 'NR>2'
fi

echo

# Alert if thresholds exceeded

if (( $(echo "$CPU_USAGE > 80" | bc -l) )) || (( $(echo "$MEM_USAGE > 80" | bc -l) )); then

    MESSAGE="High resource usage detected on $(hostname):\nCPU:
$CPU_USAGE%\nMemory: $MEM_USAGE%"

    echo -e "$MESSAGE" | mail -s "$SUBJECT" "$EMAIL"

    echo "Alert sent to $EMAIL"
else

    echo "System resource usage is normal."
fi

```

```

root@master:~/Devops-retest/Shell_script/Q3# ./Q3.sh
==== System Information =====
Date and Time: Sat May 3 06:23:56 UTC 2025

==== CPU Usage =====
CPU Usage: 4.8%

==== Memory Usage =====
Memory Usage: 52%

==== Disk Space Usage =====
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        29G   8.5G   20G   31% /
tmpfs            3.9G   0   3.9G   0% /dev/shm
tmpfs            1.6G   2.3M   1.6G   1% /run
tmpfs            5.0M   0   5.0M   0% /run/lock
/dev/xvda16      881M   137M   683M   17% /boot
/dev/xvda15      105M   6.1M   99M    6% /boot/efi
shm              64M    0   64M   0% /run/containers/storage/overlay-containers/1e9c89144dff89aa46c87ae06ed6b131550c1b7d558f3e9931c750f544b87251/userdata/shm
shm              64M    0   64M   0% /run/containers/storage/overlay-containers/718a668521f7e93435ffcbaa553b77e94083c819f81b3c4f3f3717986deab717/userdata/shm
overlay          29G   8.5G   20G   31% /var/lib/containers/storage/overlay/bceb6746dbf6e17dc696b1068f1da514634dadcf2cd74786f6d7415bdda72a9b/merged
overlay          29G   8.5G   20G   31% /var/lib/containers/storage/overlay/0f978feaaaf20e6f4136eabe62fc20fcb1a6233ba9bde5253c53da800832f3102/merged
shm              64M    0   64M   0% /run/containers/storage/overlay-containers/b7aedcd65a4ee17b3bdf450601c9b337b6de4898986f7c68aeadd64caaf39842/userdata/shm
shm              64M    0   64M   0% /run/containers/storage/overlay-containers/3852108a01c049a79795f03057bfcddae298c2fd0f1842a6a57adc2045f1a12/userdata/shm
overlay          29G   8.5G   20G   31% /var/lib/containers/storage/overlay/a6c4d6b1e7549e7d312687cc9649c6032f27607c772c1f3e1219ba59e0324557/merged
overlay          29G   8.5G   20G   31% /var/lib/containers/storage/overlay/8e5483f7af34b4fcb75f48c8430ce6b75b2797e315dde27c76a846da01f6b39/merged
overlay          29G   8.5G   20G   31% /var/lib/containers/storage/overlay/6ec276fc45c0ed1c8428bfe2fb87dd28c6ec48f7ede2ec4b58285af78f68dd6d/merged

tor: 2721149785 14192614 0 0 0 0 0 0 0 0 2721149785 14192614 0 0 0 0 0 0 0 0
enX0: 2048872866 1456743 0 0 0 0 0 0 0 0 144995494 168793 0 0 0 0 0 0 0 0
docker0: 116829917 6223 0 0 0 0 0 0 0 0 1337497 7236 0 0 0 0 0 0 0 0
datapath: 7148 122 0 0 0 0 0 0 0 0 2616 36 0 0 0 0 0 0 0 0
weave: 14438581 179621 0 0 0 0 0 0 0 0 146 17789197 192497 0 0 0 0 0 0 0 0
vethwe-datapath: 9110 125 0 0 0 0 0 0 0 0 0 2972 40 0 0 0 0 0 0 0
vethwe-bridge: 2972 40 0 0 0 0 0 0 0 0 0 9110 125 0 0 0 0 0 0 0
vxlan-6784: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
vethwepladd3175: 8488911 90042 0 0 0 0 0 0 0 0 8886666 96153 0 0 0 0 0 0 0 0
vethwepl20a9748: 8458964 89505 0 0 0 0 0 0 0 0 8922791 96628 0 0 0 0 0 0 0 0
veth848e48b: 116916997 6222 0 0 0 0 0 0 0 0 1340015 7271 0 0 0 0 0 0 0 0
vethweplb1b6145: 2518 35 0 0 0 0 0 0 0 0 0 7024 98 0 0 0 0 0 0 0

System resource usage is normal.
root@master:~/Devops-retest/Shell_script/Q3#

```